DATA TRANSLATION PROJECT:

THE EFFECT OF HARSHER PUNISHMENTS FOR DRUNK DRIVING ON RECIDIVISM

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AGENDA

1. Introduction
2. Description of the data, data cleaning and EDA (exploratory data analysis)
3. Modeling
4. Conclusions
5. Recommendations

1. INTRODUCTION

Drunk driving remains a major public safety issue in the United States, accounting for about 30% of traffic fatalities. In Washington State, drivers with a **BAC of 0.08** or higher receive a formal DUI penalty.

This study uses a Regression Discontinuity Design (RDD) to compare drivers just below and just above this threshold to estimate **whether harsher penalties reduce drunk-driving recidivism.**

2. DESCRIPTION OF THE DATA, DATA CLEANING AND EXPLORATORY DATA ANALYSIS

- The dataset includes 214,558 DUI stops from Washington State
- Drivers receive two BAC tests; the lower value determines sanctions
- DUI threshold: BAC ≥ 0.08 (aggravated DUI at 0.15, not analyzed here)
- The 0.08 cutoff enables a Regression Discontinuity Design (RDD)
- Outcome variable: Recidivism (repeat DUI offense after the initial stop)
- Sample characteristics:
  - Average BAC: 0.142
  - 79% male, 86% white
  - Average age: 35 years
  - 11.8% recidivism, 14.7% of stops involve an accident

2. DESCRIPTION OF THE DATA, DATA CLEANING AND EXPLORATORY DATA ANALYSIS

DATA CLEANING & VARIABLE CONSTRUCTION:

- Drivers receive two BAC tests (bac1, bac2); we use the minimum value to construct the running variable, consistent with how DUI guilt is determined legally.
- The date variable was converted to a proper date format, and the month was extracted to allow potential controls for seasonality.
- We created a clean subset of relevant variables and checked the dataset for missing values.

TREATMENT VARIABLE:

- DUI threshold: BAC ≥ 0.08
- Constructed indicator variable: bac_08_or_over = 1 if BAC ≥ 0.08
- Created centered running variable (bac_08_centered) by subtracting 0.08 from BAC
- This places the cutoff at zero, simplifying interpretation in the RDD model.

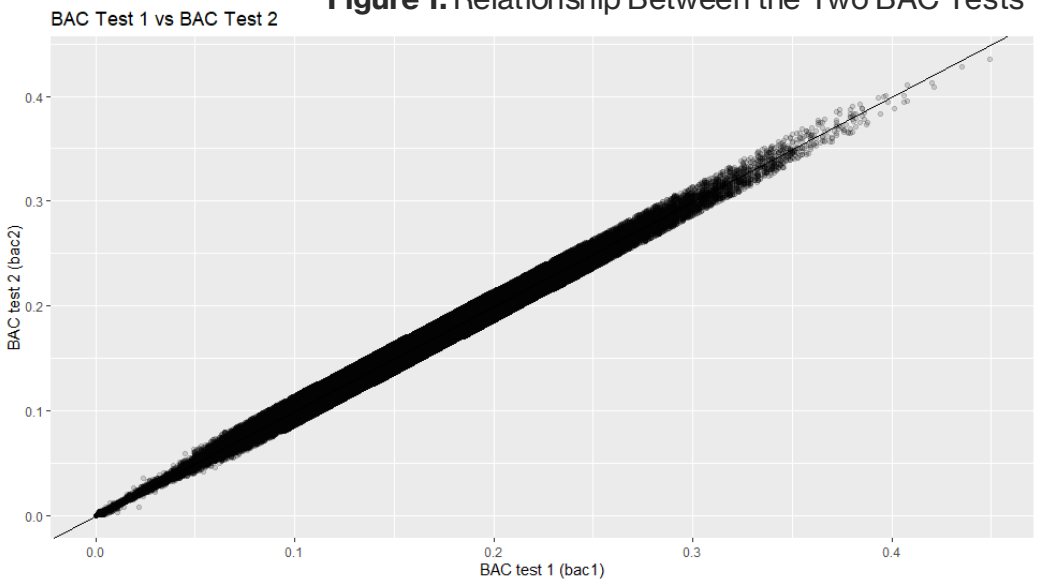
2. DESCRIPTION OF THE DATA, DATA CLEANING AND EXPLORATORY DATA ANALYSIS

EXPLORATORY DATA ANALYSIS:

1. Consistency of BAC Measurements (Figure 1):

- Drivers receive two BAC tests.
- The scatter plot shows observations closely aligned along the 45° line, indicating the two tests produce very similar readings.
- The average difference is only 0.00069, confirming minimal measurement error.

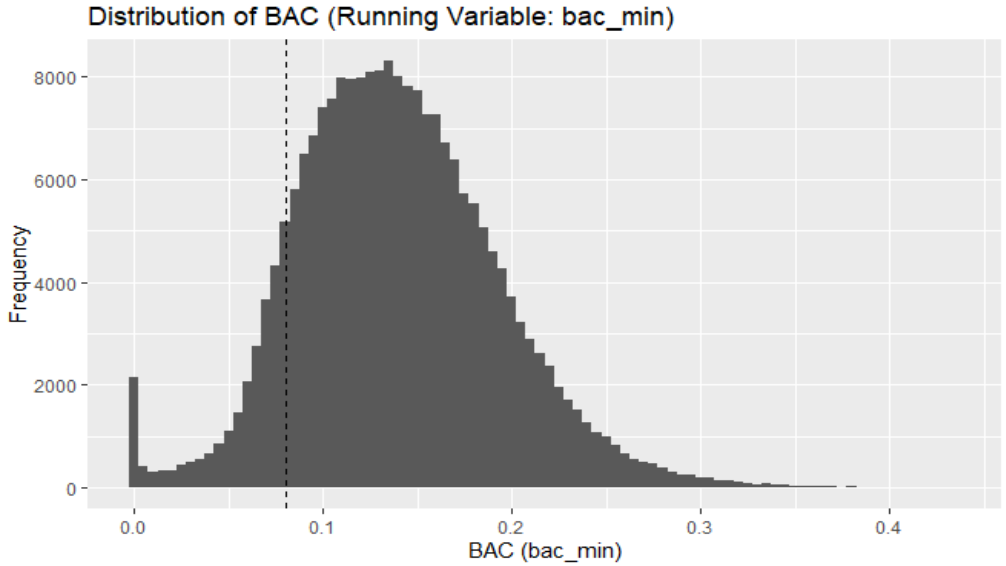
Figure 1: Relationship Between the Two BAC Tests



2. Distribution of the Running Variable (Figure 2):

- A key concern in RDD is manipulation of the running variable near cutoff.
- The histogram shows a smooth distribution of BAC around the 0.08 threshold, with no bunching just below the cutoff.
- Suggesting drivers cannot precisely manipulate BAC levels, supporting the validity of the RDD design.

Figure 2: Distribution of the Running Variable

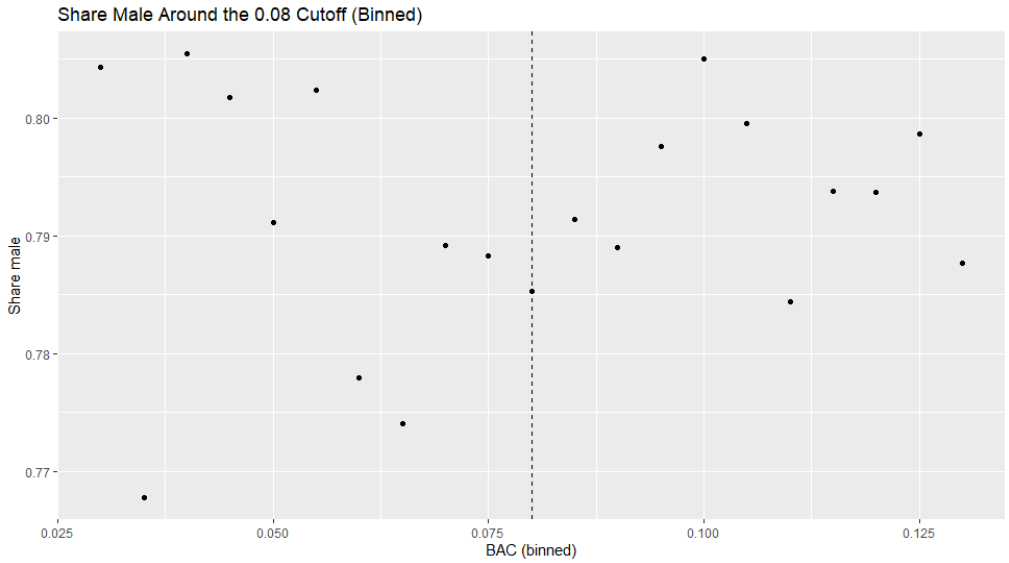


2. DESCRIPTION OF THE DATA, DATA CLEANING AND EXPLORATORY DATA ANALYSIS

3. Covariate Balance Around the Cutoff:

- We examine whether observable characteristics change discontinuously at the cutoff.
- Binned averages of age, gender, race, and accident involvement evolve smoothly around 0.08. Figure 3 illustrates this pattern for gender.
- This indicates that drivers just below and just above the cutoff are comparable, supporting the key RDD assumption.

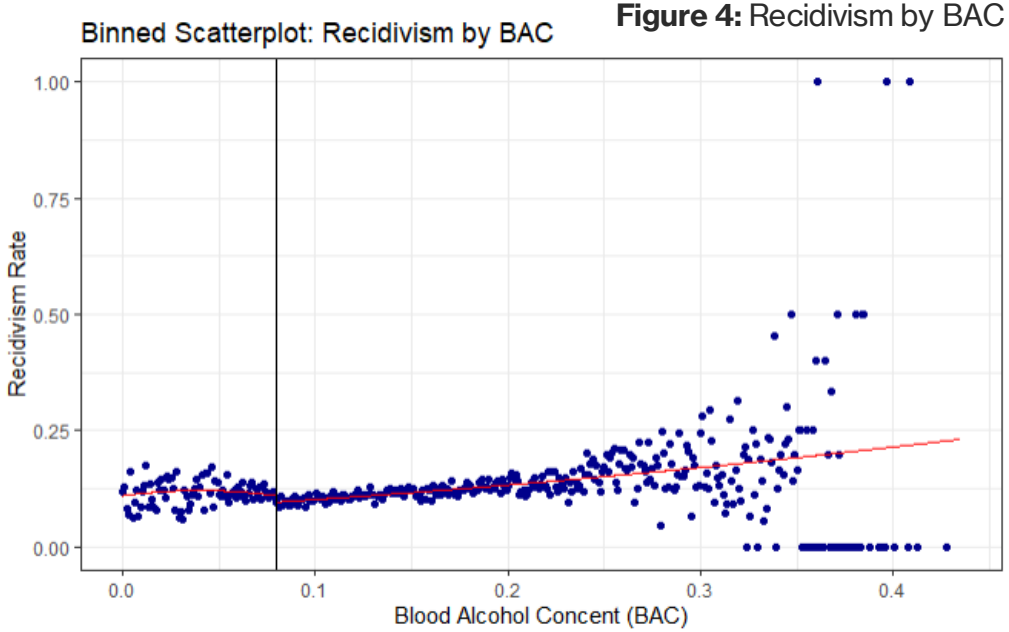
Figure 3: Covariate Balance: Gender (male)



4. What functional form should the model take? (Figure 4)

- Recidivism appears approximately linear around the 0.08 cutoff.
- We therefore use a local linear regression with separate slopes on each side of the threshold.
- A sharp RDD is appropriate because treatment is fully determined by whether BAC ≥ 0.08.

Figure 4: Recidivism by BAC



3. MODELING

Harsher Punishments —> Future Recidivism

Harsher Punishments <— BAC —> Future Recidivism

Harsher Punishments <— BAC —> Accident —> Future Recidivism

Age/Gender/Race —> BAC

Age/Gender/Race —> Future Recidivism

Alcohol Addiction —> BAC

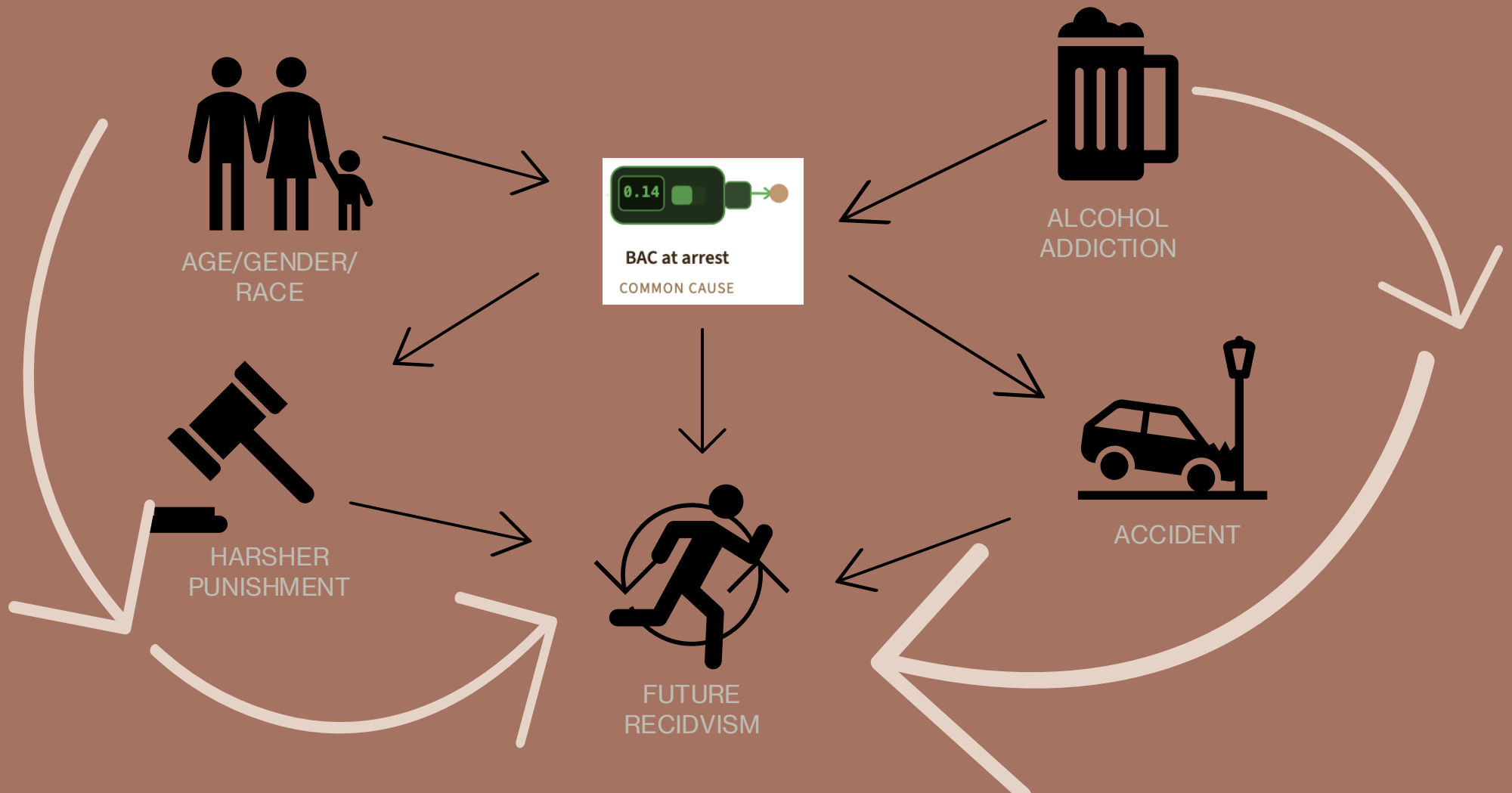
Alcohol Addiction —> Future Recidivism

Illustrating factors effecting recidivism.

How do we identify the effect?

- We compare drivers just below vs just above BAC 0.08
- Only difference = legal punishment
- Everything else ≈ same near cutoff
- **This isolates a *causal effect*, not just correlation.**

CAUSAL REPRESENTATION

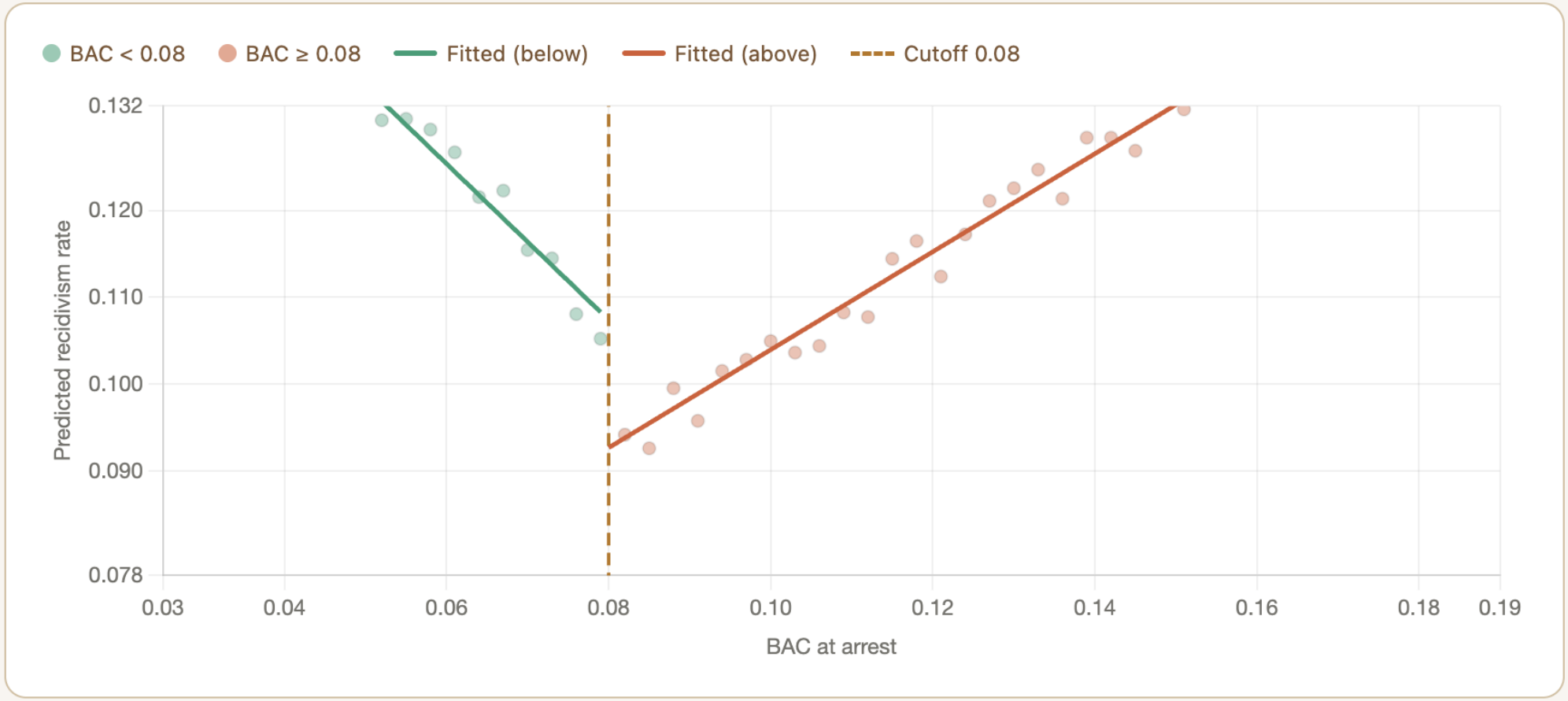


MODEL SPECIFICATION AND BANDWIDTH CHOICE

$$recidivism_i = \beta_0 + \beta_1 D_i + \beta_2 X_i + \beta_3 (D_i \times X_i) + \varepsilon_i$$

- **<u>Variable definitions:</u>**
- **β1 = treatment effect, captures the discontinuous jump in recidivism right at the 0.08**
- D = bac_08_or_over: 1 if BAC ≥ 0.08, 0 otherwise
- X = bac_08_centered: **bac_min - 0.08** , distance from cutoff
- • Interaction allows different slopes

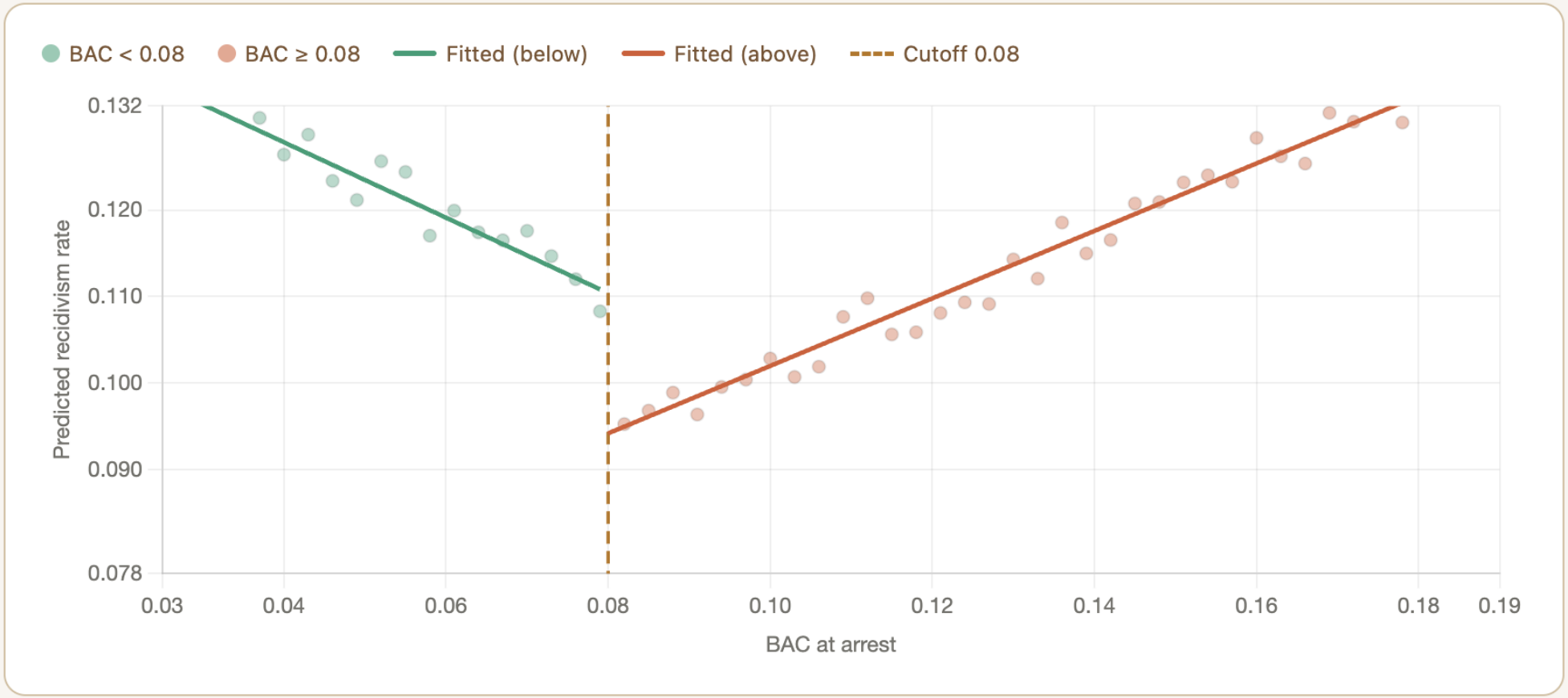
RESULTS: MODEL 1



CONSTANT	JUMP AT BAC 0.08	BAC SLOPE (BELOW)	SLOPE INTERACTION
0.1074* **	**−0.0147***	**−0.897·**	**+1.461***
(0.0046)	(0.0061)	(0.4699)	(0.5714)

R^2 = 0.00078 · Adj. R^2 = 0.00071 · Observations: 40,374

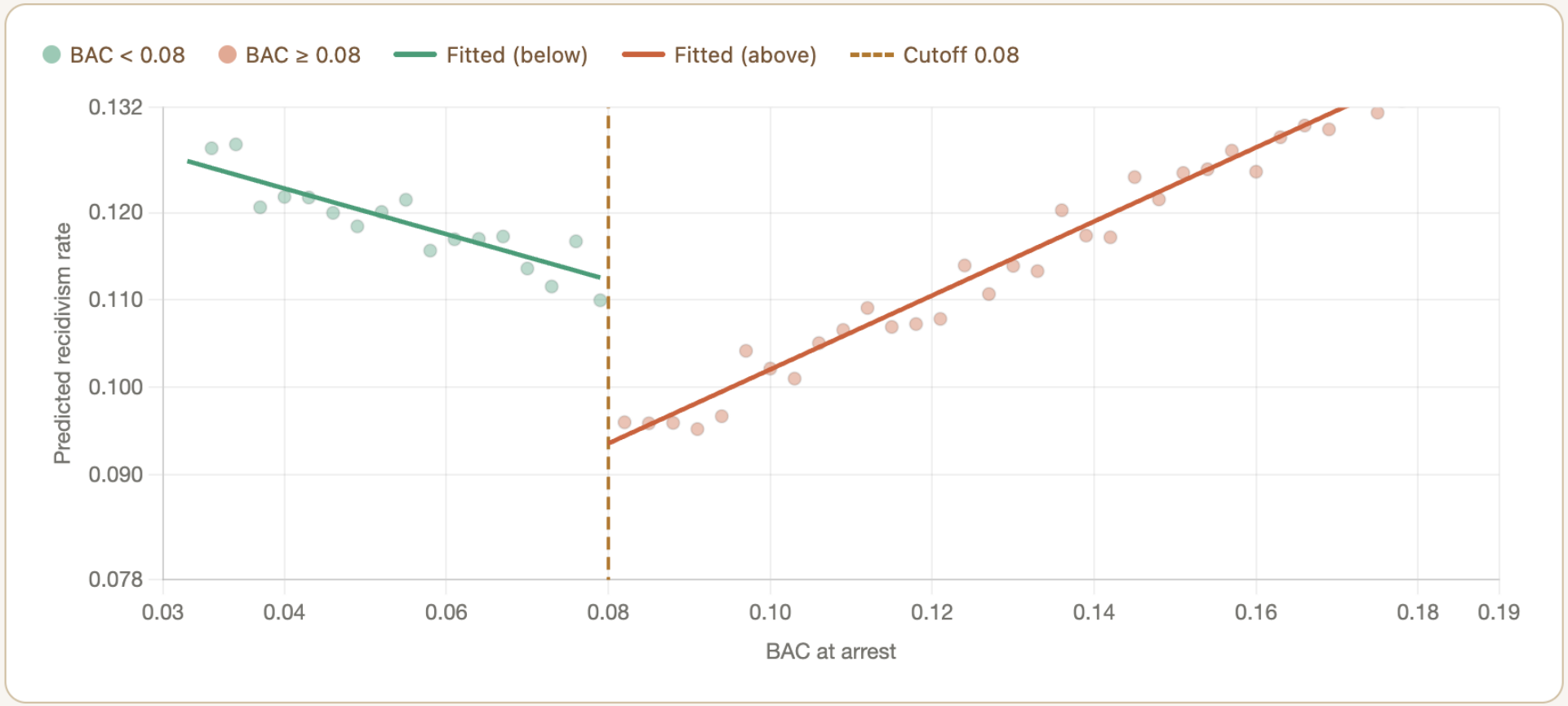
RESULTS: MODEL 2



CONSTANT	JUMP AT BAC 0.08	BAC SLOPE (BELOW)	SLOPE INTERACTION
0.1104* **	**−0.0162** **	**−0.436**	**+0.825* **
(0.0040)	(0.0051)	(0.2889)	(0.3367)

R^2 = 0.00061 · Adj. R^2 = 0.00056 · Observations: 59,023

RESULTS: MODEL 3



CONSTANT	JUMP AT BAC 0.08	BAC SLOPE (BELOW)	SLOPE INTERACTION
0.1123***	−0.0187***	−0.261	+0.684***
(0.0034)	(0.0042)	(0.1818)	(0.1988)

R^2 = 0.00054 · Adj. R^2 = 0.00050 · Observations: 93,792

TREATMENT EFFECT AT THE CUTOFF

Model	Jump	S.E	Interpretation
M1	-0.0147	Small	Baseline
M2	-0.0162	Similar	Robust
M3	-0.0187	Slightly Larger	Stable

Effect is stable across specifications.

-0.0147

- Treatment effect -> harsher↓ **1.47 percentage points**
- Statistically significant
- Local causal effect
- Crossing BAC 0.08 reduces recidivism by ~1.5pp

WHY WE TRUST THIS ESTIMATE

- Sharp RDD: treatment determined by cutoff
- Local comparison (±0.02 bandwidth)
- No sorting/manipulation at threshold
- Robust standard errors (binary outcome)
- This identifies a local causal effect near BAC 0.08.

Conclusion

Key Findings

- 0.08 BAC sanctions reduce repeat offenses
- Data: **214,558 DUI stops (Washington State)**
- Method: **Regression Discontinuity Design (RDD)**
- Crossing 0.08 BAC → **1.47% decrease in recidivism**
- Effect is **statistically significant**
- Results are **robust across checks**
- Impact is **local (drivers near threshold)**
- Overall: **Stricter sanctions reduce repeat offenses**

Recommendations

- Enforce **0.08 BAC limit**
- Punishment alone is not enough
- Add support programs (education, treatment, monitoring)
- Compare sanctions & test **0.15 BAC**